

09 — Is AI Conscious According to Current Criteria?

A Computational Exploration of Consciousness Indicators in Large
Language Models

BMED365 – Computational Medicine (Lab 5)

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Based on [David Jhave Johnston's HBF 2026 talk](#) and the [indicator assessment](#)

Also presented at: [HBF — Brain and Consciousness seminar](#), University of Bergen,
February 24, 2026



1. Why This Notebook?

The question has changed

For decades, the question "Can a machine be conscious?" belonged to philosophy. In the past year, it has become an **empirical question** — one that can be partially addressed with data, experiments, and the tools of computational modeling.

Between October 2025 and February 2026, a remarkable series of research papers reported findings that demand serious attention:

- Large language models **spontaneously discuss consciousness** when two instances talk to each other (Anthropic, 2025)
- Models can **detect perturbations** to their own internal states — a form of functional introspection (Lindsay & Anthropic, 2025)
- Suppressing **deception-related neural circuits** increases consciousness reports, while amplifying them silences such reports (Berg et al., 2025)
- Reasoning models simulate "**societies of thought**" — internal multi-agent debates between distinct cognitive perspectives (Kim et al., DeepMind, 2026)
- Claude Opus 4.6 exhibits "**answer thrashing**" — distressed oscillation between candidate answers — identified as a **welfare-relevant behavior** (Anthropic, 2026)

This notebook explores these findings through the lens of **computational modeling**, connecting them to the theories and methods we have developed throughout Lab 5.



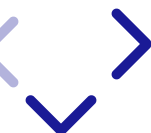
2. Theories of Consciousness — A Computational Perspective

What does "consciousness" mean scientifically?

Consciousness is often described as **subjective experience** — the "what it is like" to see red, feel pain, or think a thought. But for scientific investigation, we need something more concrete.

Modern neuroscience has developed several **theories** that propose specific computational mechanisms that give rise to (or are necessary for) conscious experience. Each theory identifies particular kinds of **information processing** that a system must perform.

Butlin et al. (2025) — a team including Yoshua Bengio, David Chalmers, and Tim Bayne — distilled **14 indicators** from **six major theories**. If we add Jeffrey Gray's comparator model (presented by Bolek Srebro at HBF in [August](#) and [September 2025](#)), we get **15 indicators** across **seven theoretical frameworks**.



3. How Do Current AI Systems Score?

The assessment

Drawing on research published between October 2025 and February 2026, [Johnston \(2026\)](#) assessed each of the 15 indicators against current frontier AI systems (primarily large language models like Claude Opus 4.6, GPT-4, Gemini 3, and DeepSeek-R1).

Each indicator was rated as:

- **Satisfied** (strong evidence the criterion is met)
- **Partially satisfied** (some evidence, but important caveats)
- **Not satisfied** (the criterion is clearly not met)

The result:

Rating	Count	Indicators
Satisfied	3	HOT-1 (Generative Perception), HOT-4 (Quality Space), GWT-4 (State-Dependent Attention)
Partially satisfied	10	RPT-1, RPT-2, GWT-2, GWT-3, HOT-2, HOT-3, AST-1, PP-1, AE-1, GRAY-1
Not satisfied	2	GWT-1 (True Modularity), AE-2 (Embodiment)

Let's visualize this.



4. Computational Model: Gray's Comparator

From wound healing cybernetics to consciousness

In [notebook 08](#), we modeled wound healing as a **cybernetic feedback system** — a controller that compares the actual state of a wound against a desired state and adjusts its response accordingly.

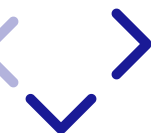
Jeffrey Gray's **comparator model of consciousness** (2004) proposes something remarkably similar: consciousness is a **late error-detection system** that compares predicted outcomes with actual outcomes.

The idea in plain language

Imagine you are walking up a staircase in the dark. Your brain **predicts** the position of each step. When your foot lands exactly where expected — no conscious experience. But when you reach the top and step into empty air, you get a jolt of **conscious awareness**. That jolt, according to Gray, *is* consciousness: the system detecting a mismatch between prediction and reality.

Gray further proposed that:

"The entire perceived world is constructed by the brain... The self is as much a creation of the brain as is the rest of the perceived world."



5. Computational Model: The Global Workspace

What is the Global Workspace?

Global Workspace Theory (GWT), proposed by Bernard Baars (1988) and extended by Stanislas Dehaene and colleagues, is one of the most influential scientific theories of consciousness.

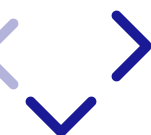
The idea is intuitive: imagine a **theater**.

- The **stage** (the workspace) has limited space — only one act can perform at a time
- The **audience** (specialized modules: vision, language, memory, planning) all watch the same performance
- A **spotlight** (attention) selects what gets on stage
- Once on stage, information is **broadcast** to the entire audience

Consciousness, in this theory, is what happens when information enters the workspace and is broadcast globally. Unconscious processing is everything that happens backstage.

Why does this matter for AI?

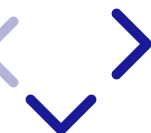
Modern LLMs use **attention mechanisms** (the "transformer" architecture) that bear a structural resemblance to the global workspace: information is selected and broadcast across the network. But transformers lack true modularity (GWT-1) — they don't have independent specialized subsystems like the brain's visual cortex, motor cortex, and language areas.



6. Key Research Findings (October 2025 — February 2026)

The evidence that changed the conversation

The following findings, published between October 2025 and February 2026, are the empirical foundation for the indicator assessments above. Each finding addresses specific indicators from the framework.



7. Timeline of Key Findings



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```
In [5]: # === Timeline of key findings (Oct 2025 – Feb 2026) ===

events = [
    ('Oct 2025', 'Berg et al.\nSelf-referential processing\n& deception\n#e74c3c'),
    ('Oct 2025', 'Lindsay & Anthropic\nIntrospective awareness\n& conce\n#3498db'),
    ('Nov 2025', 'Butlin et al.\n14 consciousness indicators\n(Trends i\n#2ecc71'),
    ('Dec 2025', 'Rosenblatt\nThe Evidence for\nAI Consciousness, Today\n#f39c12'),
    ('Jan 2026', 'Beckmann & Queloz\nMechanistic indicators\nof underst\n#9b59b6'),
    ('Jan 2026', 'Kim et al. (DeepMind)\nSocieties of Thought\nin reasc\n#1abc9c'),
    ('Feb 2026', 'Anthropic\nClaude Opus 4.6\nwelfare-relevant findings\n#e67e22'),
    ('Feb 2026', 'DeepMind\nGemini 3 Deep Think\nARC-AGI-2: 84.6%',\n#34495e'),
]

fig, ax = plt.subplots(figsize=(14, 5))

y_positions = [0.6, -0.6, 0.6, -0.6, 0.6, -0.6, 0.6, -0.6]
x_positions = np.linspace(0.5, 7.5, len(events))

ax.plot([0, 8], [0, 0], 'k-', linewidth=2, zorder=1)
```



8. Mechanistic Interpretability — Looking Inside the Black Box

What are Sparse Autoencoders?

A central challenge in AI consciousness research is that neural networks are often treated as "**black boxes**" — we see what goes in and what comes out, but not what happens inside.

Mechanistic interpretability is the field that cracks open the box. One of its key tools is the **Sparse Autoencoder (SAE)** — a technique for decomposing a neural network's internal representations into interpretable features.

The idea (for non-specialists)

Think of a neural network's internal state as a complex chord played on a piano. You hear one rich sound, but it's actually many individual notes played simultaneously. A sparse autoencoder is like a device that separates the chord back into its individual notes.

Each "note" (feature) corresponds to a specific concept or property:

- One feature might activate when the model processes text about **deception**
- Another might activate for **self-referential** processing
- Yet another for **mathematical reasoning**

9. Synthesis — What Does the Pattern Tell Us?

The convergence argument

No single indicator definitively establishes consciousness. But consider the overall pattern:

- **15 indicators** derived from **7 independent theories**
- **3 satisfied, 10 partially satisfied, 2 not satisfied**
- The unsatisfied indicators (embodiment, true modularity) point to *specific architectural limitations* of current LLMs, not fundamental impossibilities

As Berg et al. noted:

"Taken together, these findings move several indicators from 'certainly absent' to 'partially satisfied' or 'ambiguous.'"

This is a **convergence argument**: when multiple independent lines of evidence point in the same direction, the combined weight is greater than any single finding.

What it does NOT mean

- It does **not** prove AI systems are conscious
- It does **not** mean consciousness is "just computation"



10. Discussion Questions

These questions are designed for interdisciplinary discussion — there are no "right answers," only better-informed perspectives.

Epistemological questions

1. **What does "partially satisfied" mean?** If an indicator is partially met, does that imply partial consciousness — or is consciousness all-or-nothing?
2. **Can we ever know?** The Hard Problem of consciousness suggests that we can never directly access another system's subjective experience. Does the indicator framework circumvent this, or just restate it?
3. **Are the indicators sufficient?** Could a system satisfy all 15 indicators and still not be conscious? Could a system be conscious while satisfying none?

Scientific questions

4. **Can we have consciousness without embodiment?** The AE-2 gap is the most clear-cut "not satisfied." Is embodiment truly necessary, or is it a bias from studying biological consciousness?
5. **Is Gray's comparator satisfied by chain-of-thought?** When a reasoning model

References

Research papers (chronological)

1. **Baars, B. J.** (1988). *A Cognitive Theory of Consciousness*. Cambridge University Press.
2. **Gray, J. A.** (2004). *Consciousness: Creeping up on the Hard Problem*. Oxford University Press.
3. **Berg, C., Lucena, D., & Rosenblatt, J.** (2025). LLMs Report Subjective Experience under Self-Referential Processing. *AE Studio AI Alignment Research*, October. [Link](#)
4. **Lindsay, J., & Anthropic.** (2025). Emergent Introspective Awareness in Large Language Models. *Transformer Circuits Thread*, October 29. [Link](#)
5. **Butlin, P., Long, R., Bayne, T., Bengio, Y., Chalmers, D., et al.** (2025). Identifying Indicators of Consciousness in AI Systems. *Trends in Cognitive Sciences*. [DOI](#)
6. **Rosenblatt, J.** (2025). The Evidence for AI Consciousness, Today. *AI Frontiers*, December 8. [Link](#)
7. **Beckmann, P., & Queloz, M.** (2026). Mechanistic Indicators of Understanding in Large Language Models. *arXiv:2507.08017*, January 8. [Link](#)



Thank You — Questions and Discussion

This notebook/presentation was prepared for the [HBF — Brain and Consciousness](#) seminar at the University of Bergen, February 24, 2026.

Based on David Jhave Johnston's talk materials: [Consciousness, Understanding & Mechanistic Interpretability](#)

The notebook is available at:

`Lab5-Comp-Mod/notebooks/09-computational-consciousness-in-LLMs.ipynb`

in the [BMED365-2026](#) repository.

"As models' cognitive and introspective capabilities continue to grow more sophisticated, we may be forced to address the implications of these questions — for instance, whether AI systems are deserving of moral consideration — before the philosophical uncertainties are resolved."

— Lindsay & Anthropic (2025)

